UWI COMPUTING SOCIETY • TECHNICAL WORKING GROUP

DevSphere Hub: Web Architecture & Execution Plan

Central Engineering Roadmap for the Web Development Division

DIVISION LEAD	DOCUMENT TARGET	TARGET RELEASE	STATUS
Web Dev Lead	**Core Society Hub**	**Spring / Fall Cycle**	**Approved Execution Draft**

Executive Mandate: The Web Development team holds the operational anchor of DevSphere: The Computing Innovation Hub. Our mission is to engineer the responsive web portal, host member identities, unify the three sibling divisions (AI, Game Dev, Data Science), and deliver an industry-grade educational ecosystem for both general students and society members.

1. HIGH-LEVEL TECHNICAL ARCHITECTURE

To support high performance, accessibility, and zero-downtime integration across autonomous teams, the platform employs a modular full-stack decoupled architecture. The frontend handles responsive UX and dynamic portals, backed by typed API contracts and scalable persistence.

DEVSPHERE SYSTEM TOPOLOGY

CLIENT TIER	SERVICE API	STORAGE & ORM	INTEGRATION BUS
Next.js & Tailwind	**Node.js / Express**	**PostgreSQL / Prisma**	**REST & WebSockets**
SSR/SSG, Responsive UI, Dashboard Views & Auth Context	JWT Auth, RBAC Engine, Gateway & Webhook Consumer	Relational Models, Audit Logs, Asset CDN & User Schemas	Live Telemetry, AI Embeddings & Game State Endpoints

2. KEY DELIVERABLE MODULES

Module Name	Functional Scope & Responsibilities	Primary Beneficiary
Auth & Profiles	Role-based authentication (Student, Member, Admin, Executive), GitHub OAuth, personal tech badges, and portfolio links.	ALL STUDENTS
Interactive Portals	Dedicated hubs: Game Arcade (WebGL embed), AI Assistant (chat/stream), and Society Analytics visualizer.	SUB-TEAMS & PUBLIC
Project Showcase	Member directory with live demo URLs, repository links, peer-voting mechanisms, and annual hackathon submissions.	STUDENT RESUMES
Collab & Mentorship	"Find-a-Team" directory matching developers by language preference, mentor booking, and club challenge signups.	SOCIETY GROWTH

<1.2s	100%	4 Tiers	99.5%
TARGET LCP TIME	MOBILE RESPONSIVE	GRANULAR RBAC	PLATFORM UPTIME

3. MEMBER SKILL-TIER ALLOCATION MATRIX

To fulfill our society's core pedagogical mission, work packages are divided into three distinct execution tracks. Every member receives hands-on, portfolio-worthy experience under the mentorship of senior student engineers.

Beginner Track

FOUNDATIONAL UI & COMPONENTS

- **Landing Pages:** Section mastheads, Hero banners, FAQ accordions.
- **Component Library:** Buttons, cards, modals, and profile view templates.
- **Static Content:** Learning guides, club constitution page, and footer blocks.
- **Tooling Gained:** Git branching, HTML5/CSS3 semantics, Tailwind CSS utility flow.

Intermediate Track

APIS & APPLICATION STATE

- **Authentication Flow:** JWT token storage, session management, OAuth callback.
- **Dynamic Portals:** Showcase submission forms with cloud image uploads.
- **Filter & Query:** Search bar logic for projects, resource tagging system.
- **Tooling Gained:** React hooks, REST client architecture, SQL/Prisma basics.

Advanced Track

PLATFORM CORE & INFRA

- **Micro-Gateways:** Webhook pipelines for Game scores & Data Science ingestion.
- **Security & RBAC:** Rate limiting, SQL injection shields, CSRF protection.
- **CI/CD & Hosting:** Dockerization, automated GitHub Actions, staging deploys.
- **Tooling Gained:** Production DevOps, distributed systems, caching patterns.

4. CROSS-DIVISION TECHNICAL CONTRACTS

The Web Division is the master integration layer. Clear technical contracts eliminate bottlenecks and ensure the sibling groups operate autonomously without integration collisions.

Collaborating Group	Integration Mechanism & Interface	Web Dev Division Deliverable
Game Development Interactive Learning	HTML5 / WebGL Canvas sandbox (iframe or WebAssembly container). Standardized JSON telemetry for player progress.	Custom Arcade View with fullscreen modal, embedded gamepad inputs, and automated high-score dispatch to Club Leaderboard.
Artificial Intelligence Smart Assistant & Search	Streaming REST / Server-Sent Events (SSE) from FastAPI endpoint. Dynamic prompt-context payload containing user skill profile.	Real-time streaming Chat UI drawer, smart resource recommendation carousels, and natural-language search bar on resource page.
Data Science Analytics & Insights	Anonymized event logging stream (`POST /api/telemetry/event`). BI queries feeding optimized chart aggregations.	Lightweight, accessible Chart.js / Recharts dashboard components in Executive & Member portals showing skill trends.

Standardized API Contract Example: All sub-teams authenticate incoming hub telemetry via an agreed club header: `X-DevSphere-Service-Key`. Game completions broadcast `{ studentId, challengeId, runtimeMs, score }`, triggering instantaneous leaderboard mutations.

5. SEMESTER IMPLEMENTATION ROADMAP

A rigorous, four-phase sprint cycle ensures progressive delivery, continuous testing, and structured onboarding for all computing students.

Phase / Timeline	Web Dev Core Tasks	Inter-Team Sync & Joint Milestones	Review Gate
Phase 1: Weeks 1–3 FOUNDATIONS	Setup Next.js mono-repo, Tailwind theme tokens, Prisma schemas, and core routing.	Agree on REST/WebSocket schema contracts with AI, Game, and Data leads.	Git workflow training & PR template verification.
Phase 2: Weeks 4–7 CORE PORTALS	Build user registration, auth guards, dashboard layouts, and Showcase submission pages.	Embed first WebGL mini-game challenge; connect AI test endpoint via SSE stream.	Alpha internal release across executive committee.
Phase 3: Weeks 8–10 INTEGRATION	Implement Leaderboard engine, telemetry logging pipelines, and Data Science charts.	Run end-to-end integration tests with live player data and AI recommendation feed.	Campus-wide Closed Beta (Computing majors).
Phase 4: Weeks 11–12 LAUNCH & SCALE	Performance optimization, mobile bug fixes, automated backups, and production domain setup.	Host inaugural DevSphere Club Launch Hackathon; live scoreboard and showcase display.	General Public Release & Semester Handover.

6. ENGINEERING GOVERNANCE & QUALITY ASSURANCE

To prepare students for professional software environments, all contributions must satisfy standard engineering guardrails:

Version Control & Code Review

Main branch is protected. All feature code requires an approved GitHub Pull Request with at least one senior peer review. Continuous Integration (CI) enforces linting, formatting (Prettier), and build passes prior to merge.

Accessibility & Security Standards

Strict adherence to WCAG 2.1 AA color contrast standards. Passwords hashed using bcrypt (salt rounds > 10). CORS policies restricted to trusted internal service subdomains with parameterized SQL queries throughout.

7. IMMEDIATE ACTION ITEMS FOR THE LEAD

Action Item	Immediate Next Step	Target Completion
GitHub Organization	Create repository `devsphere-monorepo`, configure branch rules, and issue template tags.	End of Week 1
Sub-Group Alignment	Conduct API alignment meeting with AI, Game Dev, and Data Science group directors.	End of Week 1
Member Skill Intake	Survey web dev division members to assign Beginner, Intermediate, and Advanced roles.	Mid Week 2